Exercise: Basic Controller for the Longitudinal Motion

Given the following state-space representation, design a basic controller for the Boeing 777.

$$\begin{pmatrix} \dot{V}^* \\ \dot{\gamma} \\ \dot{\alpha} \\ \dot{q} \end{pmatrix} = \begin{bmatrix} -0.01 & -0.49 & -0.046 & -0.001 \\ 0.11 & 0.0003 & 1.14 & 0.043 \\ -0.11 & -0.0003 & -1.14 & 0.957 \\ 0.1 & 0 & -15.34 & -3.00 \end{bmatrix} \begin{pmatrix} V^* \\ \gamma \\ \alpha \\ q \end{pmatrix} + \begin{bmatrix} 0.0002 & 0.014 \\ -0.059 & 0 \\ 0.059 & 0 \\ 6.56 & 0.13 \end{bmatrix} \cdot \begin{pmatrix} \eta \\ \delta_T \end{pmatrix}$$

The controller is supposed to augment the short-period eigenmode such that:

- the ideal damping $\zeta_{SP} = \frac{1}{\sqrt{2}}$ and
- the desired eigenfrequency $\omega_{0,SP} = 4{,}5\,\frac{rad}{s}$

are achieved.

a) Analyze the eigenmode characteristics of the aircraft without a controller.
b) Propose a suitable controller structure, for which the measurements of the states in the above state space model suffice.
c) Determine the required controller gains such that the above-mentioned requirements for the short - period are fulfilled.
d) Propose an alternative control structure that accounts for inaccurate measurements of the angle of attack. Compute its required controller gains.